

HTML PRACTICE

HYPERTEXT MARK UP LANGUAGE
(Standard language for creating web pages)

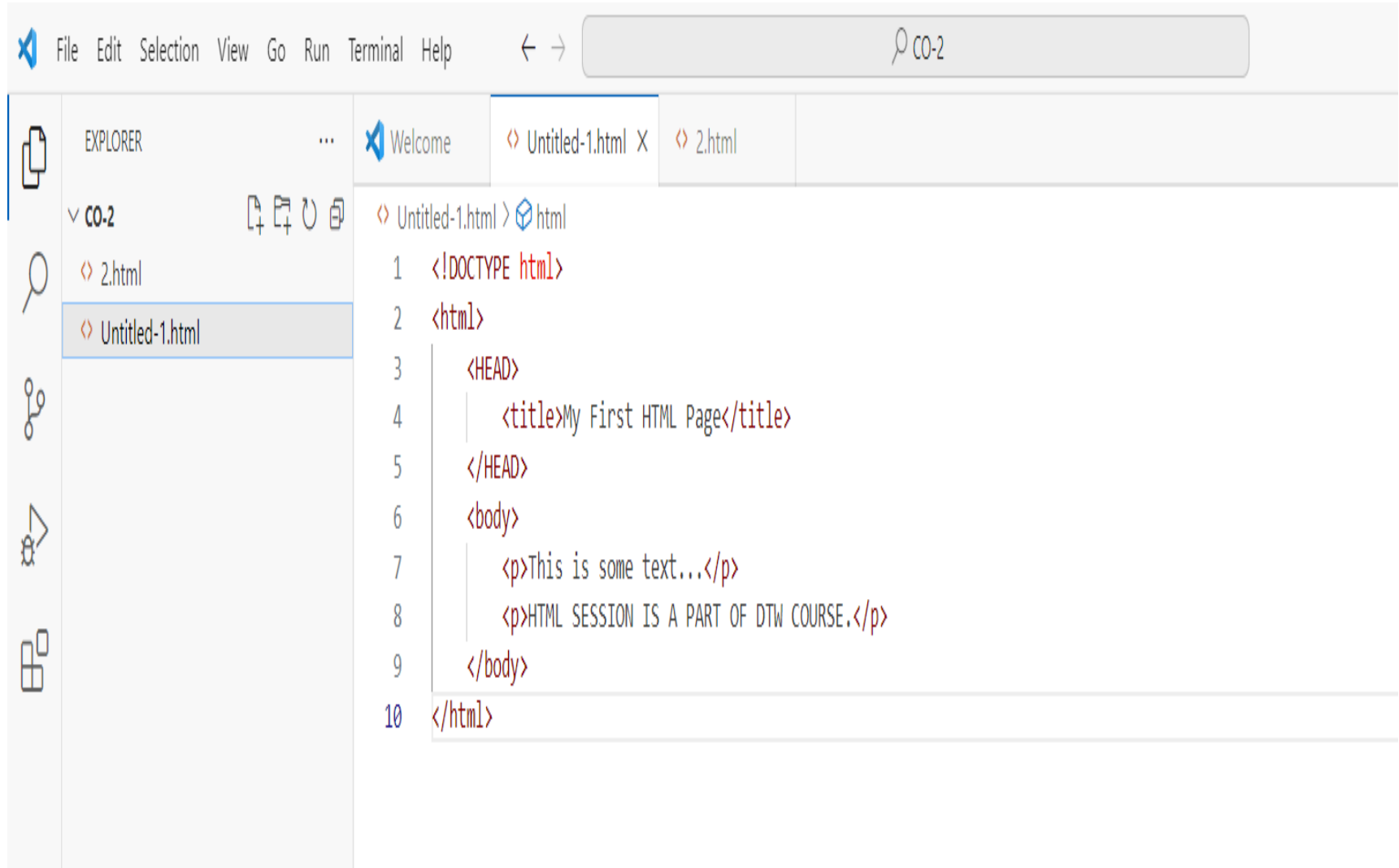
HTML BASICS

- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.
- This course follows latest HTML 5

BASIC HTML SCRIPT

```
<!DOCTYPE HTML>
<html>
  <head>
    <title>My First HTML Page</title>
  </head>
  <body>
    <p>This is some text...</p>
  </body>
</html>
```

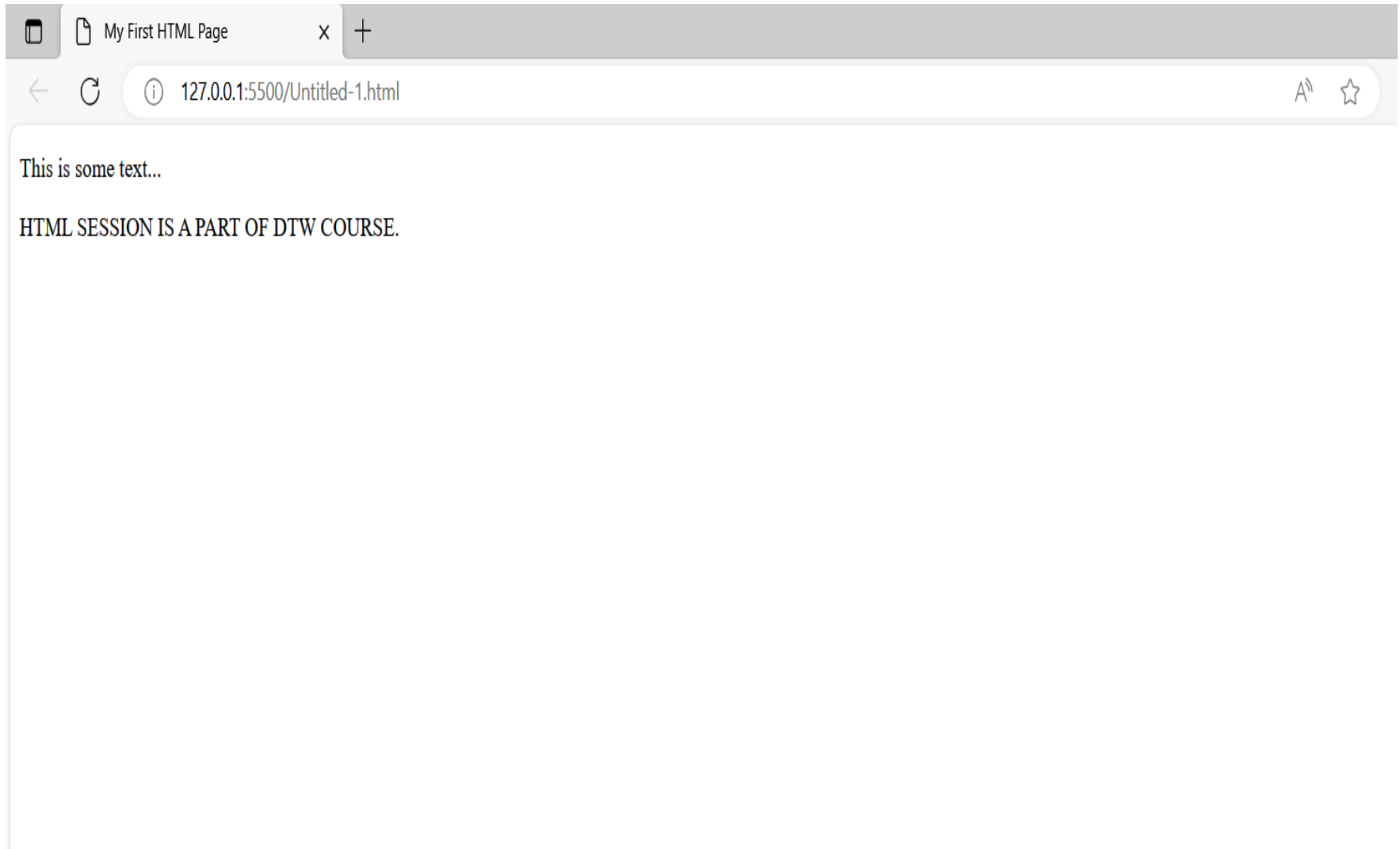
HTML SCRIPTING WITH VISUAL STUDIO CODE



Web Browsers

- The purpose of a web browser (Chrome, Edge, Firefox, Safari) is to read HTML documents and display them correctly.
- A browser does not display the HTML tags, but uses them to determine how to display the document:

OUTPUT OF BASIC HTML SCRIPT



CONTENTS OF HTML DOCUMENT

- HTML STRUCTURE CONTAINS

- ELEMENTS

- TAGS

HTML Page Structure with elements and tags

- Below is a visualization of an HTML page structure

```
<html>
```

```
<head>
```

```
<title>Page title</title>
```

```
</head>
```

```
<body>
```

```
<h1>This is a heading</h1>
```

```
<p>This is a paragraph.</p>
```

```
<p>This is another paragraph.</p>
```

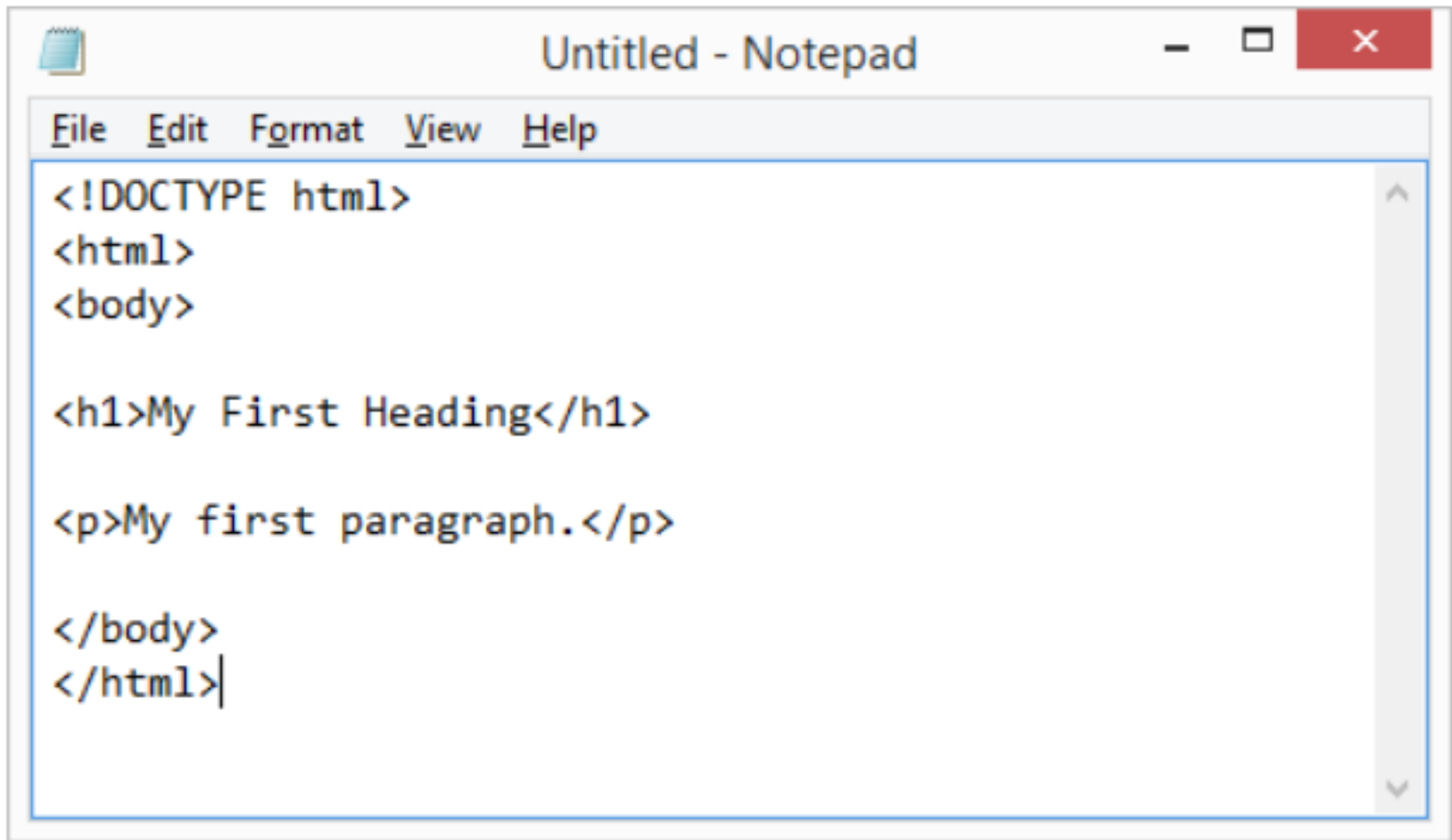
```
</body>
```

```
</html>
```


HTML EDITORS

- NOTE PAD
- TEXT EDIT (MAC)
- VISUAL STUDIO CODE

HTML WITH NOTEPAD

A screenshot of a Notepad window titled "Untitled - Notepad". The window has a standard Windows interface with a title bar, a menu bar (File, Edit, Format, View, Help), and a text area. The text area contains the following HTML code:

```
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>

<p>My first paragraph.</p>

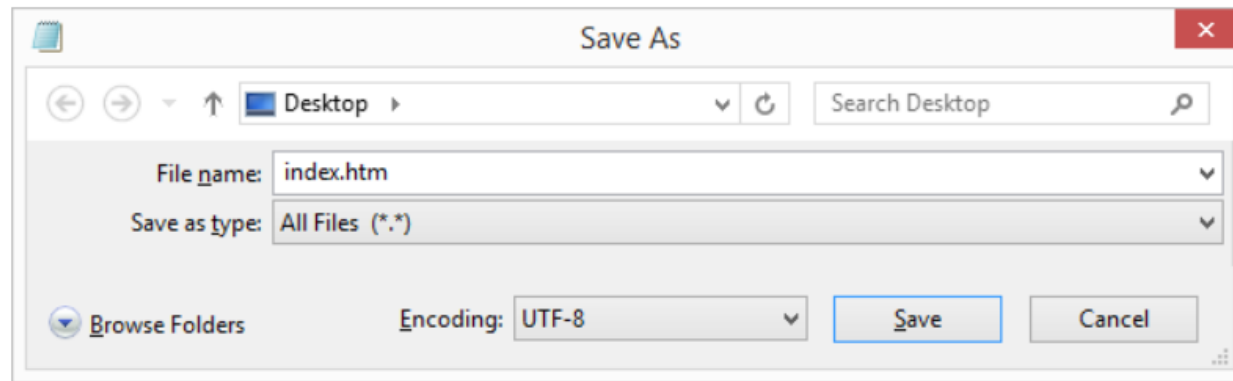
</body>
</html>
```

The cursor is positioned at the end of the last line of code. The window has a red close button in the top right corner.

SAVING HTML PAGE WITH NOTEPAD (EXTENSION .htm OR .html ARE SAME)

Save the file on your computer. Select **File > Save as** in the Notepad menu.

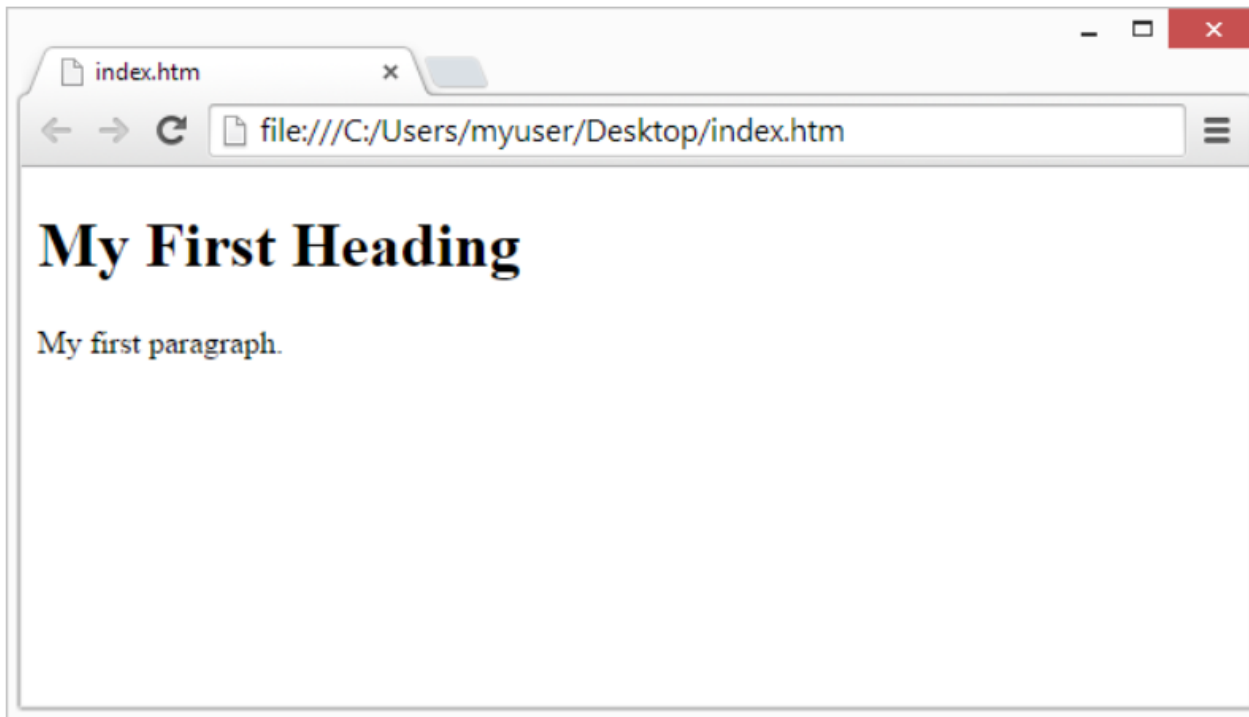
Name the file "**index.htm**" and set the encoding to **UTF-8** (which is the preferred encoding for HTML files).



VIEWING HTML PAGE IN BROWSER

Open the saved HTML file in your favorite browser (double click on the file, or right-click - and choose "Open with").

The result will look much like this:



HTML Documents

- All HTML documents must start with a document type declaration: `<!DOCTYPE html>`.
- The HTML document itself begins with `<html>` and ends with `</html>`.
- The visible part of the HTML document is between `<body>` and `</body>`.

The `<!DOCTYPE html>` Declaration

- The `<!DOCTYPE>` declaration represents the document type, and helps browsers to display web pages correctly.
- It must only appear once, at the top of the page (before any HTML tags).
- The `<!DOCTYPE>` declaration is not case sensitive.
- Declaration:
`<!DOCTYPE html>`

HTML PAGE TITLE

- Every web page should have a page title to describe the meaning of the page.
- The <title> element adds a title to your page in the browser toolbar
- The page title is very important for search engine optimization (SEO). The text is used by search engine algorithms to decide the order when listing pages in search results.

```
<head>  
  <title>HTML Tutorial</title>  
</head>
```

HTML Elements

- The HTML **element** is everything from the start tag to the end tag.
- It is defined by start tag, some content, and end tag
`<tagname>Content goes here...</tagname>`

Example:

```
<h1>My First Heading</h1>
```

```
<p>My first paragraph.</p>
```


HTML ELEMENT

Start tag	Element content	End tag
<h1>	My First Heading	</h1>
<p>	My first paragraph.	</p>
 	<i>none</i>	<i>none</i>

HTML ELEMENTS

- The content inside the `<body>` section will be displayed in a browser. The content inside the `<HEAD>` element will be shown in the browser's title bar or in the page's tab.
- Some HTML elements have no content (like the `<br>` element). These elements are called empty elements. Empty elements do not have an end tag!

Tags: HTML Headings

- HTML headings are defined with the <h1> to <h6> tags.
- <h1> defines the most important heading. <h6> defines the least important heading.

<h1>Heading 1</h1>

<h2>Heading 2</h2>

<h3>Heading 3</h3>

<h4>Heading 4</h4>

<h5>Heading 5</h5>

<h6>Heading 6</h6>

HTML Headings

- Search engines use the headings to index the structure and content of your web pages.
- Users often skim a page by its headings. It is important to use headings to show the document structure.
- `<h1>` headings should be used for main headings, followed by `<h2>` headings, then the less important `<h3>`, and so on.
- Browsers automatically add some white space (a margin) before and after a heading.
- Use HTML headings for headings only. Don't use headings to make text **BIG** or **bold**.

HTML Headings

output

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

HTML Headings

- Each HTML heading has a default size. However, you can specify the size for any heading with the style attribute, using the CSS font-size property.

```
<h1 style="font-size:60px;">Heading 1</h1>
```

TAGS: HTML paragraphs

- HTML paragraphs are defined with the `<p>` tag.
- Example:

```
<p>This is a paragraph.</p>
```

```
<p>This is another paragraph.</p>
```

TAGS: HTML links

- HTML links are defined with the `<a>` anchor tag.
- The link's destination is specified in the href attribute.
- Attributes are used to provide additional information about HTML elements.
- Example:

`<a href=https://www.klh.edu.in>This is a link</a>`

TAGS: HTML images

- HTML images are defined with the `<img>` tag.
- The source file (src), alternative text (alt), width, and height are provided as attributes:
- The source files are to be saved in the same folder/location where html script file is saved
- Example:

```

```

EXAMPLE

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h1>My First Heading</h1>
```

```
<p>My first paragraph.</p>
```

```
</body>
```

```
</html>
```

Nested HTML Elements

- HTML elements can be nested (this means that elements can contain other elements).
- All HTML documents consist of nested HTML elements.
- The following example contains four HTML elements (<html>, <body>, <h1> and <p>):

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h1>My First Heading</h1>
```

```
<p>My first paragraph.</p>
```

```
</body>
```

```
</html>
```

Nested HTML Elements in previous example

- The `<html>` element is the root element and it defines the whole HTML document.
- It has a start tag `<html>` and an end tag `</html>`.
- Then, inside the `<html>` element there is a `<body>` element:
- The `<body>` element defines the document's body.
- It has a start tag `<body>` and an end tag `</body>`.

Nested HTML Elements in previous example

- Then, inside the `<body>` element there are two other elements: `<h1>` and `<p>`:
- The `<h1>` element defines a heading.
- It has a start tag `<h1>` and an end tag `</h1>`:
- The `<p>` element defines a paragraph.
- It has a start tag `<p>` and an end tag `</p>`:
- **Never Skip the End Tag**
- HTML tags are not case sensitive: `<P>` means the same as `<p>`

HTML ATTRIBUTES

- All HTML elements can have **attributes**
- Attributes provide **additional information** about elements
- Attributes are always specified in **the start tag**
- Attributes usually come in name/value pairs like:
name="value"

href attributes

- The <a> tag defines a hyperlink. The href attribute specifies the URL of the page the link goes to.
- Example:

<a href=<https://klh.edu.in/>> Visit klh website

src Attribute

- The `<img>` tag is used to embed an image in an HTML page. The `src` attribute specifies the path to the image to be displayed.
- There are two ways to specify the URL in the `src` attribute:
- **1. Absolute URL** - Links to an external image that is hosted on another website.
In addition, you cannot control external images; it can suddenly be removed or changed.
- **2. Relative URL** - Links to an image that is hosted within the website. Here, the URL does not include the domain name.
If the URL begins without a slash, it will be relative to the current page.
Example: `src="img_girl.jpg"`.
- If the URL begins with a slash, it will be relative to the domain. Example: `src="/images/img_girl.jpg"`.

The width and height and alt Attributes

- The `<img>` tag should also contain the width and height attributes, which specify the width and height of the image (in pixels).

```

```

- **The alt Attribute** for the `<img>` tag specifies an alternate text for an image, if the image for some reason cannot be displayed.
- This can be due to a slow connection, or an error in the src attribute, or if the user uses a screen reader.

```

```

The lang Attribute

- You should always include the lang attribute inside the `<html>` tag, to declare the language of the Web page. This is meant to assist search engines and browsers.

```
<!DOCTYPE html>  
<html lang="en">  
<body>  
...  
</body>  
</html>
```

The style Attribute

- The style attribute is used to add styles to an element, such as color, font, size, and more.

```
<p style="color:red;">This is a red paragraph.</p>
```

- Double quotes around attribute values are the most common in HTML, but single quotes can also be used.
- In some situations, when the attribute value itself contains double quotes, it is necessary to use single quotes. Both the examples are correct.

```
<p title="John 'ShotGun' Nelson">
```

```
<p title='John "ShotGun" Nelson'>
```

Title attribute

- The title attribute defines some extra information about an element.

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2 title="I'm a header">The title Attribute</h2>
```

```
<p title="I'm a tooltip">Mouse over this paragraph, to display the title attribute as a tooltip.</p>
```

```
</body>
```

```
</html>
```

HTML STYLES

- The HTML style attribute is used to add styles to an element, such as color, font, size, and more.

- SYNTAX

- The *property* is a CSS property. The *value* is a CSS value.

`<tagname style="property:value;">`

HTML STYLES

- The CSS background-color property defines the background color for an HTML element.

```
<body style="background-color:powderblue;">
```

```
<h1>This is a heading</h1>
```

```
<p>This is a paragraph.</p>
```

```
</body>
```

HTML STYLES

- SET background color with two different elements

```
<body>
```

```
<h1 style="background-color:powderblue;">This is a heading</h1>
```

```
<p style="background-color:tomato;">This is a paragraph.</p>
```

```
</body>
```

HTML STYLES

- The CSS color property defines the text color for an HTML element.

```
<h1 style="color:blue;">This is a heading</h1>
```

```
<p style="color:red;">This is a paragraph.</p>
```

 ; the font to be used for an HTML element.

```
<h1 style="font-family:verdana;">This is a heading</h1>
```

```
<p style="font-family:courier;">This is a paragraph.</p>
```


HTML STYLES

- The CSS font-size property defines the text size for an HTML element.

```
<h1 style="font-size:300%;">This is a heading</h1>
```

```
<p style="font-size:160%;">This is a paragraph.</p>
```

- The CSS text-align property defines the horizontal text alignment for an HTML element.

```
<h1 style="text-align:center;">Centered Heading</h1>
```

```
<p style="text-align:center;">Centered paragraph.</p>
```

Html formatting elements

- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Smaller text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text

Html formatting elements

`<b>This text is bold</b>`

`<strong>This text is important!</strong>`

`<i>This text is italic</i>`

`<em>This text is emphasized</em>`

`<small>This is some smaller text.</small>`

`<p>My favorite color is <del>blue</del> <ins>red</ins>.</p>`

`<p>My favorite color is <del>blue</del> red.</p>`

`<p>This is <sub>subscripted</sub> text.</p>`

`<p>This is <sup>superscripted</sup> text.</p>`

`<p>Do not forget to buy <mark>milk</mark> today.</p>`

HTML COMMENT TAG

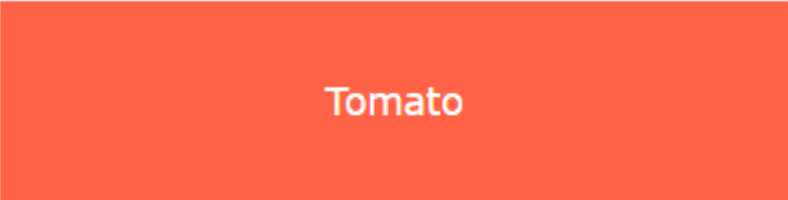
- SYNTAX:

`<!-- Write your comments here -->`

- There is an exclamation point (!) in the start tag, but not in the end tag.
- Comments can be used to hide content.
- This can be helpful if you hide content temporarily.

Html colors

In HTML, a color can be specified by using a color name:



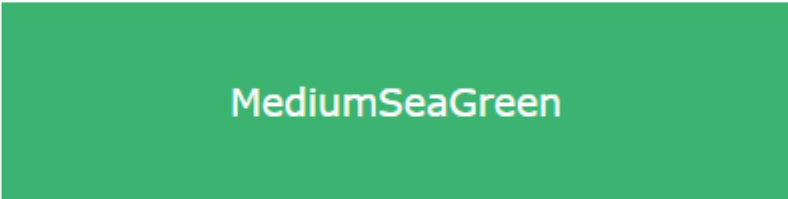
Tomato



Orange



DodgerBlue



MediumSeaGreen



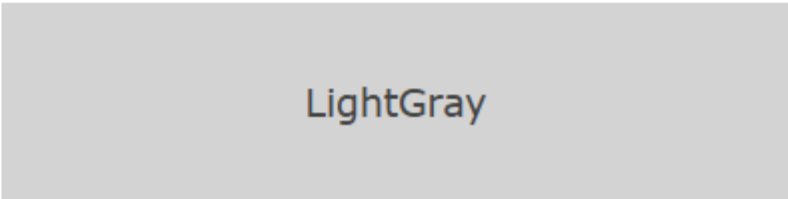
Gray



SlateBlue



Violet



LightGray

HTML IMAGES

TRY THIS EXAMPLE

```
<!DOCTYPE html>
<html>
<head>
<style>
img {
  width: 100%;
}
</style>
</head>
<body>





</body>
</html>
```

HTML COLORS

- HTML colors are specified with predefined color names, or with RGB, HEX, HSL, RGBA, or HSLA values.

```
<!TEXT COLOR>
```

```
<html>
```

```
<body>
```

```
<h1 style="background-color:DodgerBlue;">Hello World</h1>
```

```
<p style="background-color:Tomato;">
```

K L Deemed to be University, officially the Koneru Lakshmaiah Education Foundation, is a deemed university located in the Vaddeswaram Guntur district of Andhra Pradesh, offering academic programs at UG, PG, Doctoral, Post-Doctoral industry focused courses. Now K L Deemed to be University is operational in Hyderabad.

```
</p>
```

```
</body>
```

```
</html>
```

Border Color

```
<h1 style="border:2px solid Tomato;">Hello World</h1>
```

```
<h1 style="border:2px solid DodgerBlue;">Hello World</h1>
```

```
<h1 style="border:2px solid Violet;">Hello World</h1>
```


HTML TABLES

TRY OUT SIMPLE TABLE

```
<table>
  <tr>
    <th>Company</th>
    <th>Contact</th>
    <th>Country</th>
  </tr>
  <tr>
    <td>Alfreds Futterkiste</td>
    <td>Maria Anders</td>
    <td>Germany</td>
  </tr>
  <tr>
    <td>Centro comercial Moctezuma</td>
    <td>Francisco Chang</td>
    <td>Mexico</td>
  </tr>
</table>
```

Html tables

- td stands for table data.
- Everything between `<td>` and `</td>` are the content of the table cell.
- tr stands for table row.
- Each table row starts with a `<tr>` and ends with a `</tr>` tag.
- th stands for table header.
- By default, the text in `<th>` elements are bold and centered, but you can change that with CSS.

Html tables

```
<table>
  <tr>
    <th>Person 1</th>
    <th>Person 2</th>
    <th>Person 3</th>
  </tr>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
  <tr>
    <td>16</td>
    <td>14</td>
    <td>10</td>
  </tr>
</table>
```

HTML TABLE TAGS

HTML Table Tags

Tag	Description
<u><table></u>	Defines a table
<u><th></u>	Defines a header cell in a table
<u><tr></u>	Defines a row in a table
<u><td></u>	Defines a cell in a table
<u><caption></u>	Defines a table caption
<u><colgroup></u>	Specifies a group of one or more columns in a table for formatting
<u><col></u>	Specifies column properties for each column within a <colgroup> element
<u><thead></u>	Groups the header content in a table
<u><tbody></u>	Groups the body content in a table
<u><tfoot></u>	Groups the footer content in a table

HTML LISTS

- HTML lists allow web developers to group a set of related items in lists.
- An unordered list starts with the `<ul>` tag. Each list item starts with the `<li>` tag.
- The list items will be marked with bullets (small black circles) by default:
- An ordered list starts with the `<ol>` tag. Each list item starts with the `<li>` tag.
- The list items will be marked with numbers by default:
- HTML also supports description lists.
- A description list is a list of terms, with a description of each term.
- The `<dl>` tag defines the description list, the `<dt>` tag defines the term (name), and the `<dd>` tag describes each term.

TRY EXAMPLES FOR UNORDERED, ORDERED AND DESCRIPTION LISTS

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

```
<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

```
<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>
```

HTML List Tags

Tag	Description
<u></u>	Defines an unordered list
<u></u>	Defines an ordered list
<u></u>	Defines a list item
<u><dl></u>	Defines a description list
<u><dt></u>	Defines a term in a description list
<u><dd></u>	Describes the term in a description list

How to View HTML Source

- "Hey! How did they do that web page?"
- **View HTML Source Code:**
 - Right-click in an HTML page and select "View Page Source" (in Chrome) or "View Source" (in Edge), or select "Inspect" or similar in other browsers.
 - This will open a window containing the HTML source code of the page.

THANK YOU